MASt3R-SLAM: Real-Time Dense SLAM with 3D Reconstruction Priors

Imperial College London

Riku Murai*, Eric Dexheimer*, Andrew J. Davison

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Presenter: 송우석

- **site:** https://edexheim.github.io/mast3r-slam/
- **code:** https://github.com/rmurai0610/MASt3R-SLAM
- **paper:** https://arxiv.org/pdf/2412.12392

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Intro

- Visual SLAM
  - foundational building block for today's robotics and augmented reality products

  → However, SLAM is not yet a plug-and-play algorithm as it requires hardware expertise and calibration
- In-the-wild SLAM
  - provides both accurate poses and consistent dense maps for a minimal single camera setup without additional sensing such as an IMU

  → does not exist

⇒ Achieving such a reliable dense SLAM system would open new research avenues for spatial intelligence

- variety of priors
  - Single-view priors (e.g. monocular depth & normals)

  → contain ambiguities and lack consistency across views
  - Multi-view priors (e.g. optical flow)

  → challenging since pixel motion depends on both the extrinsics and the camera model
- two-view 3D reconstruction priors
  - DUSt3R, MASt3R → aforementioned subproblems are implicitly solved

Related Work

- Sparse Monocular SLAM
  - enabled real-time pose estimation and sparse reconstructions on large scale scenes

    → lacks a dense scene model which is useful for both robust tracking and more explicit reasoning over geometry
- Dense Monocular SLAM
  - recent work has attempted to combine data driven priors with backend optimisation and achieve greater consistency

    → robust correspondence across multiple views is essential
- Volumetric representations
  - adopted differentiable rendering in neural fields [25] and Gaussian splatting [18] for consistent reconstruction

    → have lagged in real-time performance compared to alternatives, and require depth, additional 2D priors, or slow camera motion

⇒ All systems mentioned thus far assume known intrinsic calibration

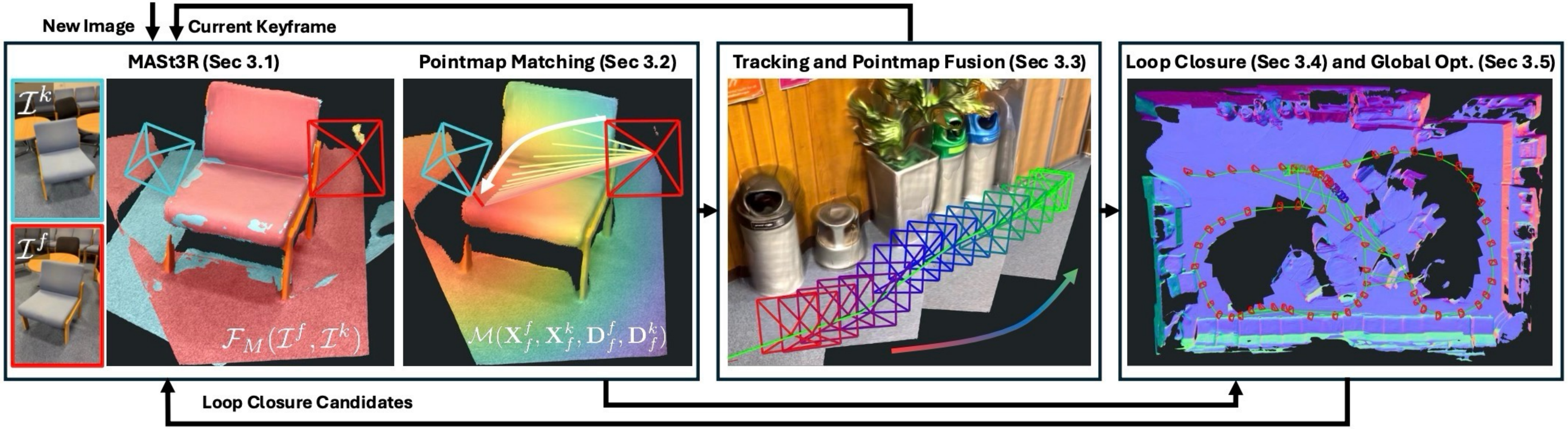
- Classical automatic intrinsic calibration

  → strict assumptions on scene geometry or unchanging parameters across a set of images (sensitive to noise)
- MASt3R

  → as with all priors, predictions can still have inconsistencies and correlated errors in the 3D geometry

MASt3R-SLAM

- a dense SLAM system built around two-view 3D reconstruction priors
- only assume a generic central camera model (all rays pass through a unique camera centre)
- propose efficient methods for pointmap matching, tracking and pointmap fusion, loop closure, and global optimisation
- achieve large scale consistency of the pairwise predictions in real-time



MASt3R-SLAM

- notation

$$\mathrm{T} = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix}, \quad \mathrm{T} \leftarrow \tau \oplus \mathrm{T} \triangleq \mathrm{Exp}(\tau) \circ \mathrm{T}, \qquad (1)$$

- $\mathbf{T} \in \mathbf{Sim}(3)$
- $\tau \in \mathfrak{sim}(3)$
- $\mathbf{R} \in \mathbf{SO}(3)$
- $\mathbf{t} \in \mathbb{R}^3$
- $s \in \mathbb{R}$

- MASt3R

$$\mathcal{F}_M(\mathcal{I}^i, \mathcal{I}^j)$$

- $\mathbf{X}_i^i, \mathbf{X}_i^j \in \mathbb{R}^{H \times W \times 3}$

  → pointmap

  - $\mathbf{X}_j^i$: the pointmap of image i represented in the coordinate frame of camera j
- $\mathbf{C}_i^i, \mathbf{C}_i^j \in \mathbb{R}^{H \times W \times 1}$

  → confidence

- additional head is added to predict
  - $\mathbf{D}_i^i, \mathbf{D}_i^j \in \mathbb{R}^{H \times W \times d}$

    → d-dimensional features for matching
  - $\mathbf{Q}_i^i, \mathbf{Q}_i^j \in \mathbb{R}^{H \times W \times 1}$

    → its corresponding confidences

MASt3R-SLAM

- Pointmap Matching
  - Correspondence = fundamental component of SLAM
    → we need to find the set of pixel matches between the two images

$$\mathrm{m}_{i,j} = \mathcal{M}(\mathrm{X}_i^i, \mathrm{X}_i^j, \mathrm{D}_i^i, \mathrm{D}_i^j)$$

MASt3R feature

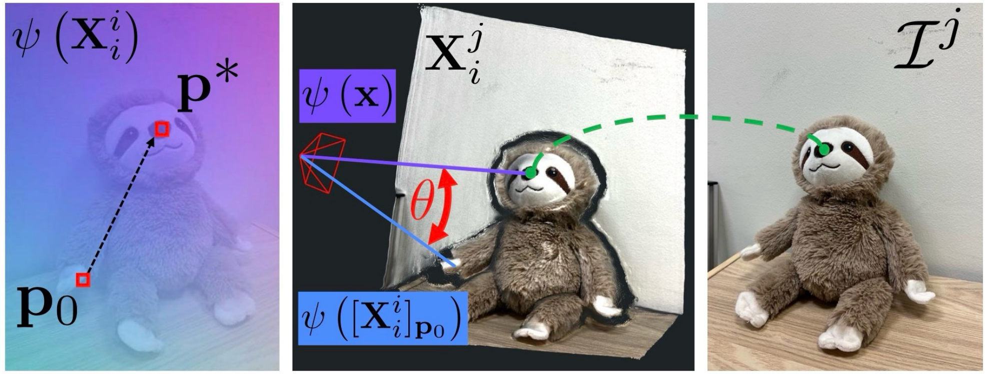
  - efficient methods for a global search → instead find inspiration from optimisation as a local search
  - motivated by the use of projective data-association commonly used in dense SLAM
    → requires a parametric camera model with closed-form projection

$$\mathbf{p}^* = \arg\min_{\mathbf{p}} \|\psi([\mathbf{X}_i^i]_{\mathbf{p}}) - \psi(\mathbf{x})\|^2. \quad (2)$$

- $\psi([\mathbf{X}_i^i]_{\mathbf{p}})$: queried ray
- $\psi(\mathbf{x})$: target ray

$\psi(\mathbf{X}_i^i)$

normalises a pointmap $\mathbf{X}_i^i$ into **rays** of unit norm such that each pointmap defines its own camera model



angle between two normalised rays

$$\|\psi_1 - \psi_2\|^2 = 2(1-\cos\theta), \quad \cos\theta = \psi_1^T\psi_2. \quad (3)$$

$\mathbf{m}_{i,j}$: initial matches

within 10 iter as ray image is smooth

- use nonlinear least-squares form
- iteratively solve for updates to projected locations by calculating analytical Jacobians and solving via Levenberg-Marquardt

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MASt3R-SLAM

- Tracking and Pointmap Fusion
  - 3D point error

$\mathcal{I}^f$ $\mathcal{I}^k$

relative transformation between the current frame and the last keyframe

$$E_p = \sum_{m,n \in \mathbf{m}_{f,k}} \left\| \frac{\tilde{\mathbf{X}}^k_{k,n} - \mathbf{T}_{kf}\mathbf{X}^f_{f,m}}{w(\mathbf{q}_{m,n}, \sigma_p^2)} \right\|_\rho , \tag{4}$$

$$\mathbf{q}_{m,n} = \sqrt{\mathbf{Q}^f_{f,m}\mathbf{Q}^k_{f,n}}$$

huber norm

match confidence weighting proposed in MASt3R-SfM [10]

While $\mathbf{X}^k_f$ instead of $\mathbf{X}^f_f$ could also be aligned to $\mathbf{X}^k_k$ with the benefit of no explicit matching required as they are pixel aligned

→ we found that explicit matching with $\mathbf{X}^f_f$ had improved accuracy for larger baseline scenarios

$$w(\mathbf{q}, \sigma^2) = \begin{cases} \sigma^2/\mathbf{q} & \mathbf{q} > \mathbf{q}_{min} \\ \infty & \text{otherwise} \end{cases} . \tag{5}$$

⇒ but, 3D point error is easily skewed by inconsistent predictions in depth (cause tracking error,degrades pointmap used in backend)

- directional ray error (less sensitive to incorrect depth predictions)

$$E_r = \sum_{m,n \in \mathbf{m}_{f,k}} \left\| \frac{\psi(\tilde{\mathbf{X}}^k_{k,n}) - \psi(\mathbf{T}_{kf}\mathbf{X}^f_{f,m})}{w(\mathbf{q}_{m,n}, \sigma_r^2)} \right\|_\rho . \tag{6}$$

MASt3R-SLAM

- Tracking and Pointmap Fusion
  - directional ray error (less sensitive to incorrect depth predictions)

$$E_r = \sum_{m,n \in \mathbf{m}_{f,k}} \left\| \frac{\psi(\tilde{\mathbf{X}}^k_{k,n}) - \psi(\mathbf{T}_{kf}\mathbf{X}^f_{f,m})}{w(\mathbf{q}_{m,n}, \sigma_r^2)} \right\|_\rho . \quad (6)$$

- angular errors are bounded → ray-based errors are robust against outliers
- include an distance error term with a small weight to prevent degeneracy from pure rotation

perturbation

$$(\mathbf{J}^T\mathbf{W}\mathbf{J})\tau = -\mathbf{J}^T\mathbf{W}\mathbf{r}, \quad \mathbf{T}_{kf} \leftarrow \tau \oplus \mathbf{T}_{kf}. \quad (7)$$

- $\mathbf{r}$: residuals
- $\mathbf{J}$: Jacobians
- $\mathbf{W}$: weights

- efficiently solve for updates to the pose using Gauss-Newton in an iteratively reweighted least-squares (IRLS) framework
- calculate analytical Jacobians of the ray and distance errors with respect to a perturbation τ of the relative pose $\mathbf{T}_{kf}$

$$\tilde{\mathbf{X}}^k_k \leftarrow \frac{\tilde{\mathbf{C}}^k_k \tilde{\mathbf{X}}^k_k + \mathbf{C}^k_f \left(\mathbf{T}_{kf}\mathbf{X}^k_f\right)}{\tilde{\mathbf{C}}^k_k + \mathbf{C}^k_f}, \tilde{\mathbf{C}}^k_k \leftarrow \tilde{\mathbf{C}}^k_k + \mathbf{C}^k_f. \quad (8)$$

	ATE [m] w/o calib	ATE [m] w/ calib
Recent	0.207	0.160
First	0.114	0.059
Median	0.102	**0.039**
Weighted	**0.097**	**0.039**

- use transform T_{kf} to update the canonical pointmap $\tilde{X}^k_k$ via a running weighted average filter
- weighted average was best method for maintaining coherence while filtering out noise

MASt3R-SLAM

- Graph Construction and Loop Closure
  - keyframe addition

    number of valid matches of $\mathbf{m}_{f,k} < w_k$ threshold

    → $\mathcal{K}_i$ added

    + bidirectional edge to the previous keyframe $\mathcal{K}_{i-1}$ is added to the edge-list $\mathcal{E}$

  - loop closure
    - adapt Aggregated Selective Match Kernel (ASMK) framework used by MASt3R-SfM [10] for image retrieval from encoded features

1. query the database with the encoded features of $\mathcal{K}_i$ to obtain the top-K images.
2. if, **retrieval scores** $> w_r$ and if, **number of matches** $> w_l$

   → give these pairs to the MASt3R decoder and add bidirectional edges
3. **update the retrieval database** by adding the new keyframe's encoded features to the inverted file index.

MASt3R-SLAM

- Backend Optimisation

$$\mathrm{T}_{ij} = \mathrm{T}_{WC_i}^{-1}\mathrm{T}_{WC_j}$$

$$E_g = \sum_{i,j\in\mathcal{E}} \sum_{m,n\in\mathbf{m}_{i,j}} \left\| \frac{\psi\left(\tilde{\mathrm{X}}_{i,m}^{i}\right) - \psi\left(\mathrm{T}_{ij}\tilde{\mathrm{X}}_{j,n}^{j}\right)}{w(\mathrm{q}_{m,n}, \sigma_r^2)} \right\|_\rho, \qquad (9)$$

  - minimize ray error for all edges $\mathcal{E}$ in the graph
  - accumulates 14×14 blocks into the $7N\times7N$ Hessian
  - use Gauss-Newton but with sparse Cholesky decomposition as the system is not dense

- Relocalisation
  - triggered if the system loses tracking due to an insufficient number of matches
  - For a new frame, the retrieval database is queried with a stricter threshold(than loop closure) on the score

    → if retrieved images have a sufficient number of matches with the current frame, added as a new keyframe into the graph and tracking resumes

- Known Calibration
  1. use backprojected pointmap by using known camera model and depth instead of canonical pointmap
  2. optimisation to be in pixel space rather than ray space

$$E_\Pi = \sum_{i,j\in\mathcal{E}} \sum_{m,n\in\mathbf{m}_{i,j}} \left\| \frac{\mathrm{p}_{i,m}^{i} - \Pi\left(\mathrm{T}_{ij}\tilde{\mathrm{X}}_{j,n}^{j}\right)}{w(\mathrm{q}_{m,n}, \sigma_\Pi^2)} \right\|_\rho, \qquad (10)$$

Experiments

- ## Datasets
  - TUM RGB-D [38]
  - 7-Scenes [36]
  - ETH3D-SLAM [34]
  - EuRoC [3]

- ## Tasks
  - Camera Pose Estimation
  - Dense Geometry Evaluation
  - Qualitative Results
  - Component Analysis (Ablation Study)

- ## Systems
  - desktop with Intel Core i9 12900K 3.50GHz
  - a single NVIDIA GeForce RTX 4090
  - runs at roughly 15 FPS (subsample every 2 frames of the datasets)
  - use the full resolution outputs from MASt3R

Camera Pose Estimation

- Datasets
  - TUM RGB-D [38]
  - 7-Scenes [36]
  - ETH3D-SLAM [34]
  - EuRoC [3]

		360	desk	desk2	floor	plant	room	rpy	teddy	xyz	avg
Calibrated	**ORB-SLAM3 [4]**	X	<u>0.017</u>	0.210	X	0.034	X	X	X	**0.009**	-
	DeepV2D [42]	0.243	0.166	0.379	1.653	0.203	0.246	0.105	0.316	0.064	0.375
	DeepFactors [6]	0.159	0.170	0.253	0.169	0.305	0.364	0.043	0.601	0.035	0.233
	DPV-SLAM [22]	0.112	0.018	0.029	0.057	0.021	0.330	0.030	0.084	<u>0.010</u>	0.076
	DPV-SLAM++ [22]	0.132	0.018	0.029	0.050	0.022	0.096	0.032	0.098	<u>0.010</u>	0.054
	GO-SLAM [54]	0.089	**0.016**	<u>0.028</u>	<u>0.025</u>	0.026	<u>0.052</u>	**0.019**	0.048	<u>0.010</u>	<u>0.035</u>
	DROID-SLAM [45]	0.111	0.018	0.042	**0.021**	**0.016**	**0.049**	<u>0.026</u>	0.048	0.012	0.038
	Ours	**0.049**	**0.016**	**0.024**	<u>0.025</u>	<u>0.020</u>	0.061	0.027	**0.041**	**0.009**	**0.030**
Uncalibrated	**DROID-SLAM* [45, 48]**	0.202	0.032	0.091	0.064	0.045	0.918	0.056	<u>0.045</u>	0.012	0.158
	Ours*	<u>0.070</u>	0.035	0.055	0.056	0.035	0.118	0.041	0.114	0.020	0.060

Table 1. Absolute trajectory error (ATE (m)) on TUM RGB-D [38].

- Metric
  - ATE RMSE (scaled trajectory alignment)

calibrates the intrinsics using Geo-Calib [48] on the first image of a sequence

	chess	fire	heads	office	pumpkin	kitchen	stairs	avg
NICER-SLAM	**0.033**	0.069	0.042	0.108	0.200	**0.039**	0.108	0.086
DROID-SLAM	<u>0.036</u>	<u>0.027</u>	<u>0.025</u>	**0.066**	0.127	<u>0.040</u>	<u>0.026</u>	<u>0.049</u>
Ours	0.053	**0.025**	**0.015**	<u>0.097</u>	**0.088**	0.041	**0.011**	**0.047**
Ours*	0.063	0.046	0.029	0.103	<u>0.114</u>	0.074	0.032	0.066

Table 2. Absolute trajectory error (ATE (m)) on 7-Scenes [36].

- Results
  - TUM RGB-D
    - Ours achieves SoTA trajectory error
    - comparable to results from recent learned techniques such as DPV-SLAM
  - 7-Scenes
    - Ours* outperforms NICER-SLAM, which uses multiple priors in depth, normal, and optical flow networks and runs offline

Camera Pose Estimation

- Datasets
  - TUM RGB-D [38]
  - 7-Scenes [36]
  - ETH3D-SLAM [34]
  - EuRoC [3]
- Metric
  - ATE RMSE (scaled trajectory alignment)
- Results
  - ETH3D-SLAM
    - no subsampling (due to fast camera motion)
    - longer tail in terms of robustness resulting in both the best ATE and AUC
  - EuRoC
    - used undistorted images on Ours*, but underperformed than DROID-SLAM (because explicitly augments its training with 10% greyscale images)

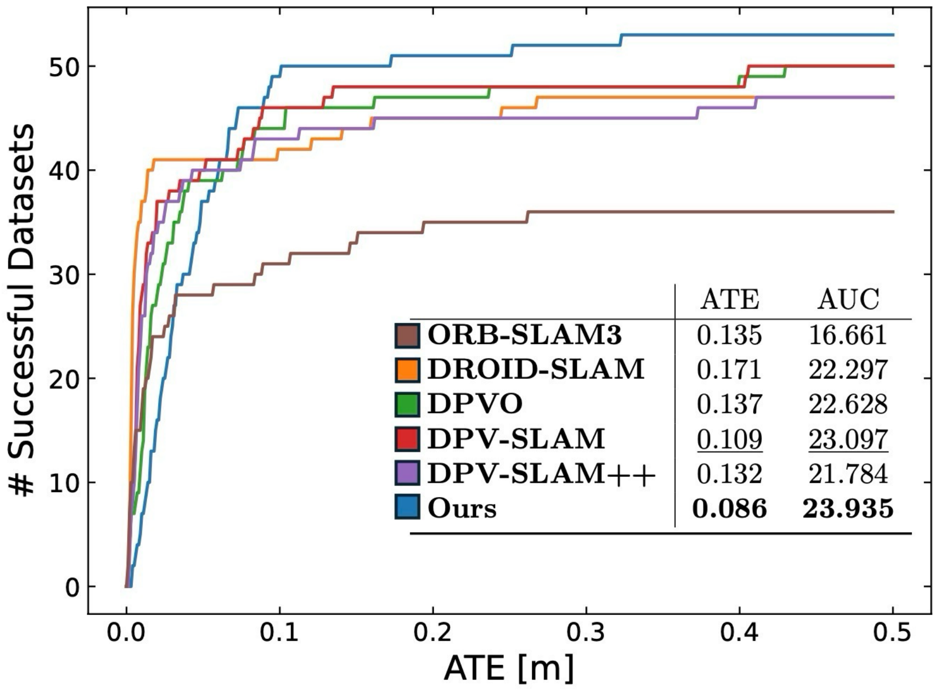


Figure 5. Number of successful trajectories below ATE threshold on ETH3D-SLAM (train) benchmark.

		MH01	MH02	MH03	MH04	MH05	V101	V102	V103	V201	V202	V203	avg
Calibrated	**ORB-SLAM**	0.071	0.067	0.071	0.082	0.060	**0.015**	0.020	X	0.021	0.018	X	-
	DeepV2D [42]	0.739	1.144	0.752	1.492	1.567	0.981	0.801	1.570	0.290	2.202	2.743	1.298
	DeepFactors [6]	1.587	1.479	3.139	5.331	4.002	1.520	0.679	0.900	0.876	1.905	1.021	2.040
	DPV-SLAM [22]	**0.013**	0.016	0.022	0.043	**0.041**	0.035	**0.008**	**0.015**	0.020	0.011	0.040	0.024
	DPV-SLAM++ [22]	**0.013**	0.016	**0.021**	**0.041**	**0.041**	0.035	0.010	**0.015**	0.021	0.011	0.023	0.023
	GO-SLAM [54]	0.016	0.014	0.023	0.045	0.045	0.037	0.011	0.023	**0.016**	**0.010**	0.022	0.024
	DROID-SLAM [45]	**0.013**	**0.012**	0.022	0.048	0.044	0.037	0.013	0.019	0.017	**0.010**	**0.013**	**0.022**
	Ours	0.023	0.017	0.057	0.113	0.067	0.040	0.019	0.027	0.020	0.025	0.043	0.041
Uncalibrated	**Ours***	0.180	0.124	0.156	0.282	0.327	0.101	0.134	0.096	0.133	0.100	0.170	0.164

Table 9. Absolute trajectory error (ATE (m)) on EuRoC [3].

Dense Geometry Evaluation

- ## Datasets
  - 7-Scenes (seq-01)
  - EuRoC (Vicon room sequences)

- ## Metric
  - accuracy (RMSE)
    - distance between each estimated point and its nearest reference point
  - completion (RMSE)
    - the distance between each reference point and its nearest estimated point
  - Chamfer Distance
    - average of the accuarcy and completion

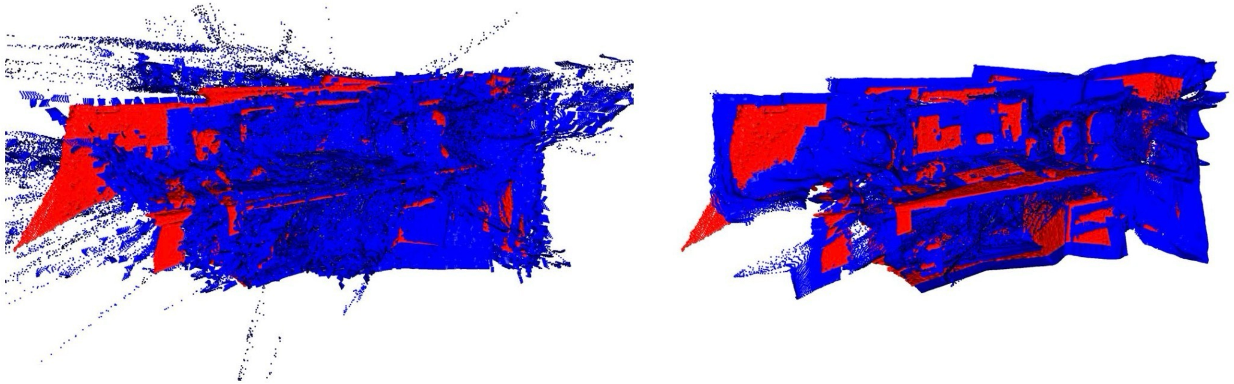
- ## Results
  - 7-Scenes
    - Ours* performs the best in both Accuracy and Chamfer distance (intrinsic calibration 7-Scenes provides is the default factory calibration)
  - EuRoC
    - Ours* has a noticeably larger ATE, but still outperforms DROID-SLAM in Chamfer distance

7-scenes	ATE	Accuracy	Completion	Chamfer
DROID-SLAM	0.049	0.115	**0.040**	0.077
Spann3R @20	N/A	0.069	0.047	0.058
Spann3R @2	N/A	0.124	0.043	0.084
Ours	**0.047**	0.074	0.057	0.066
Ours*	0.066	**0.068**	0.045	**0.056**

Table 3. Reconstruction Evaluation on 7-Scenes with all metrics in metres.

EuRoC	ATE	Accuracy	Completion	Chamfer
DROID-SLAM	**0.022**	0.173	**0.061**	0.117
Ours	0.041	**0.099**	0.071	**0.085**
Ours*	0.164	0.108	0.072	0.090

Table 3. Reconstruction Evaluation on EuRoC with all metrics in metres.



DROID-SLAM
Mean Chamfer: 0.0154m
RMSE Chamfer: 0.0604m

Ours
Mean Chamfer: 0.0142m
RMSE Chamfer: 0.0288m

Figure 9. Reconstruction comparison on 7-Scenes
(**red:** GT point cloud / **blue:** estimated point cloud)

Qualitative Results

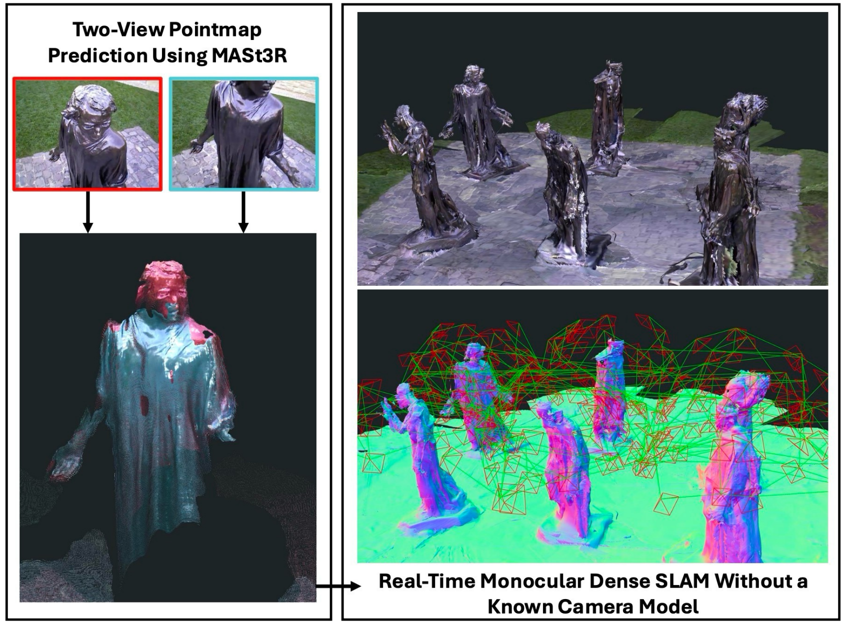


Figure 1. Reconstruction from our dense monocular SLAM system on the Burghers sequence.

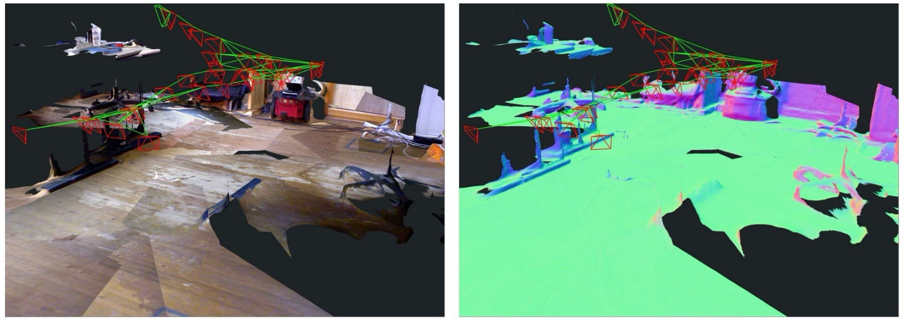


Figure 4. Reconstruction and trajectory TUM fr1/floor sequence.

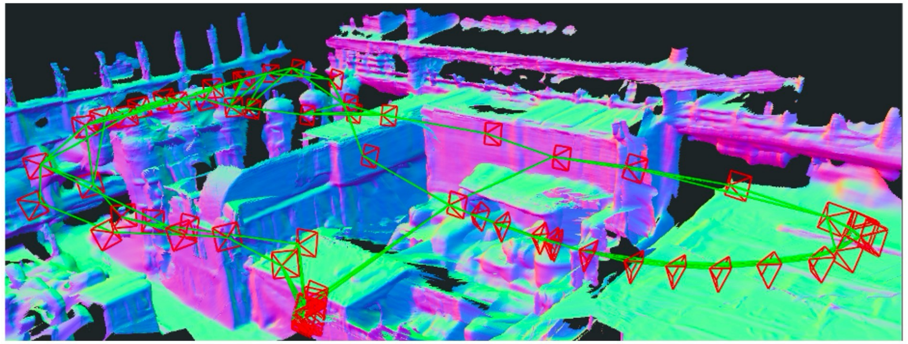


Figure 6. Reconstruction on EuRoC Machine Hall 04.

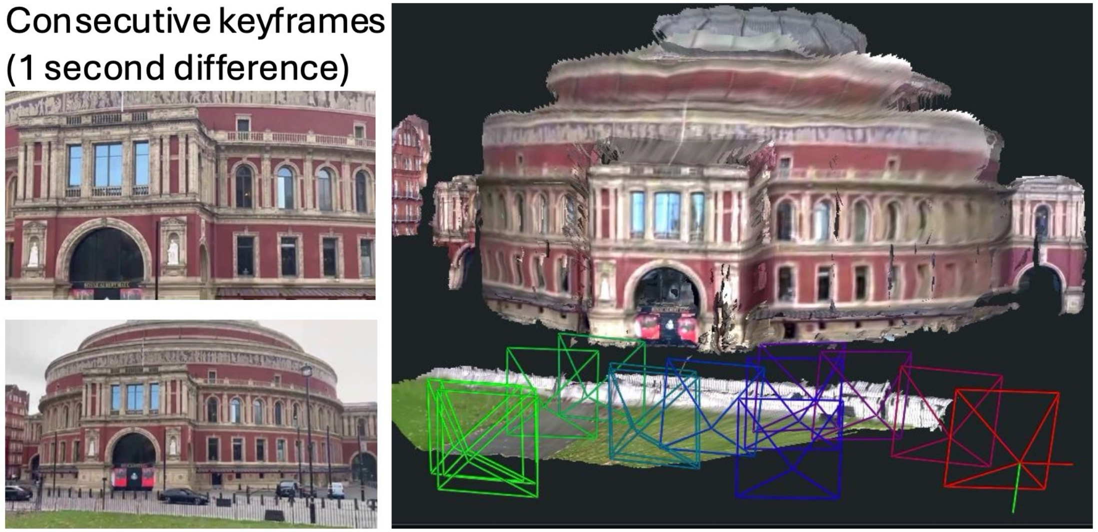


Figure 7. Dense uncalibrated SLAM with extreme zoom changes shown by two consecutive keyframes for an outdoor scene.

Component Analysis (Ablation Study)

	ATE [m] w/ calib	ATE [m] w/o calib	Matching Time [ms]	System FPS
k-d tree	0.061	0.115	40	8.8
MASt3R	0.042	0.098	2000	0.4
Ours	0.062	**0.092**	**0.5**	**15.1**
Ours + feat	**0.039**	0.097	2	14.9

Table 4. Matching comparison.

	ATE [m] w/o calib	ATE [m] w/ calib
Recent	0.207	0.160
First	0.114	0.059
Median	0.102	**0.039**
Weighted	**0.097**	**0.039**

Table 5. Fusion methods.

TUM	7-Scenes	EuRoC	avg
0.092 / **0.060**	0.084 / **0.066**	0.290 / **0.164**	0.155 / **0.097**

Table 6. ATE (m) for error formulation in format (point / ray).

	ATE (m)			Chamfer (m)
	TUM	7-scenes	EuRoC Vicon	EuRoC Vicon
Calib	0.064 / **0.030**	0.066 / **0.047**	0.233 / **0.029**	0.151 / **0.085**
No Calib	0.090 / **0.060**	0.075 / **0.066**	0.349 / **0.122**	0.179 / **0.090**

Table 7. Loop closure ablation in format (without LC / with LC).

Conclusion

- real-time dense SLAM system based on MASt3R that handles in-the-wild videos
- different approach against DROID-SLAM by building a system around an off-the-shelf geometric prior while also providing consistent dense geometry

- Limitation
  - do not currently refine all geometry in the full global optimisation
  - MASt3R's geometry predictions degrade with increasing distortion because it is only trained on images with pinhole images
  - using the decoder at full resolution is currently a bottleneck, especially for low-latency tracking and checking loop closure candidates

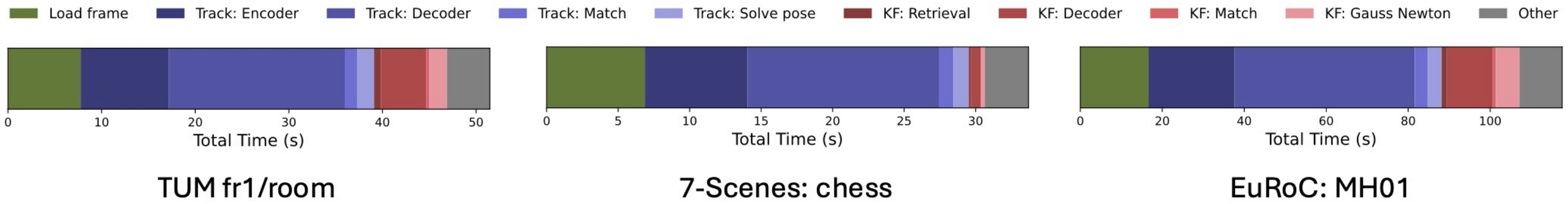


Figure 8. Total runtime in seconds for representative datasets showing cumulative time spent in significant components.